第六讲 圆轴扭转平面假设与扭转应力

作业：4-1(b),(c),6,10,11,12,13

第四章 扭转

4.1 扭力偶矩与扭矩

4.2 圆轴扭转应力

4.3 圆轴扭转强度条件与合理强度设计

4.4 圆轴扭转变形与刚度条件

4.5 圆轴扭转静不定

4.6 非圆截面杆件扭转

4.7 薄壁杆扭转

上讲回顾

拉压静不定

- 综合考虑几何、物理、静力学三方面；
- 画好变形图，看清变形协调条件；
- 看清温度和制造偏差导致的杆长变化。

静不定结构的受力特征

内力按刚度分配

变形协调，能者多劳

如何理解？

第六讲 圆轴扭转平面假设与扭转应力

作业：4-1(b),(c),6,10,11,12,13

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4.6 非圆截面杆件扭转

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第四章 扭转

4.1 扭力偶矩与扭矩

1、扭转实例与特征

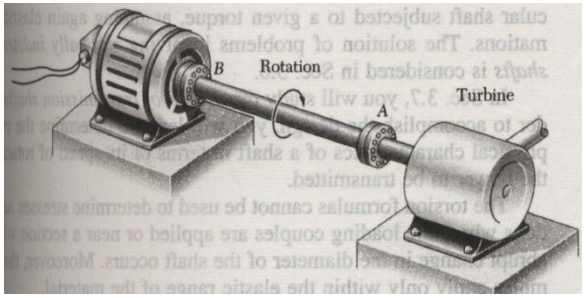
2、轴的动力传递

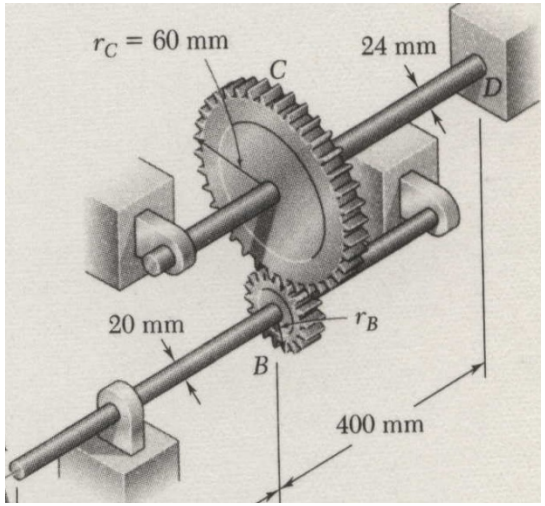
3、扭矩与扭矩图

4、例题

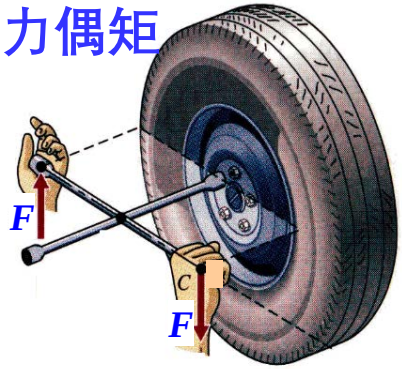
1、扭转实例与特征

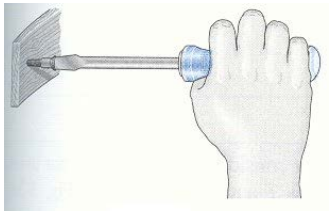
轴和齿轮传递的力矩



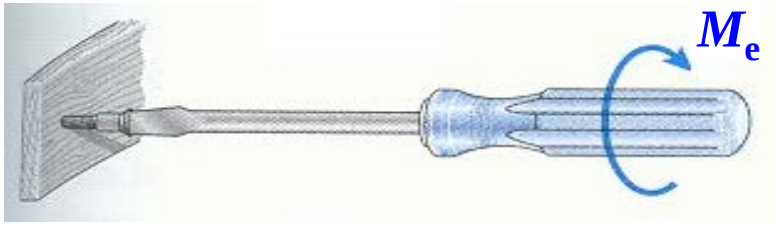


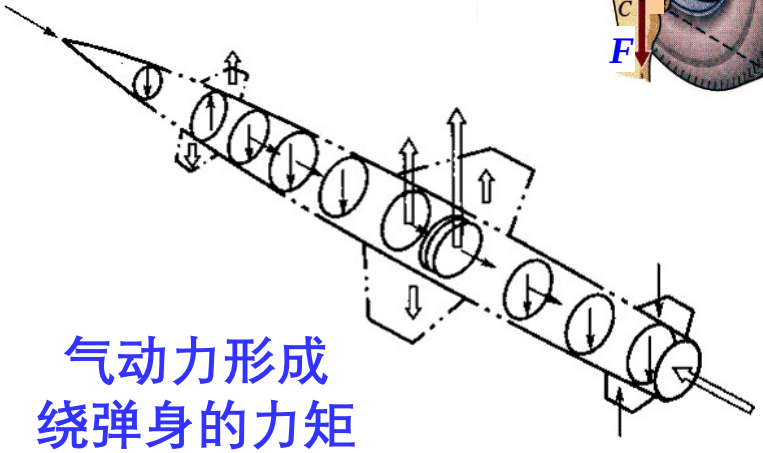
力偶矩





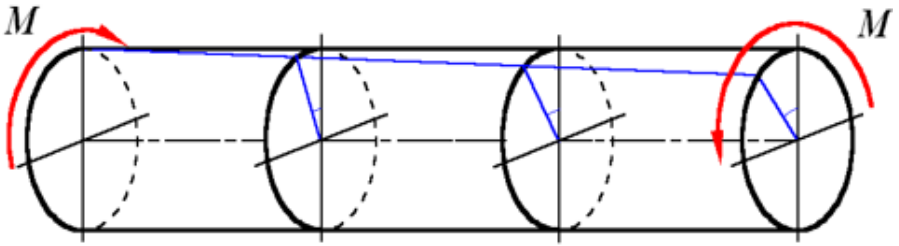
分布的摩擦力
形成力矩





气动力形成
绕弹身的力矩

1、扭转实例与特征



外力：绕轴线的力矩

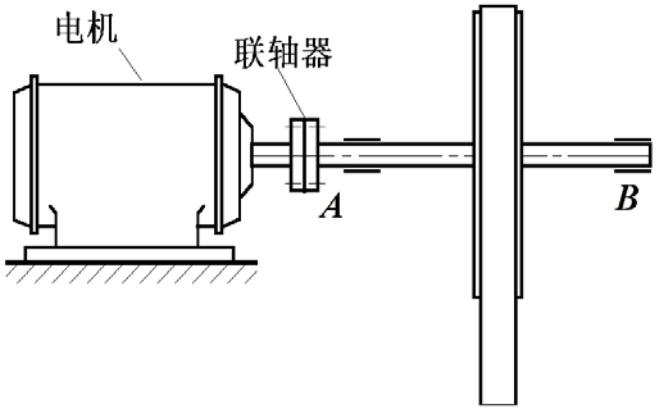
（扭力偶矩、扭力矩）

- 扭转变形：各横截面绕轴线相对转动，扭转角。
- 圆轴：受扭圆截面直杆。

2、轴的动力传递

已知：动力装置的输出功率 P（kW），转速 n（r/min）

试求：传递给轴的扭力偶矩 M（N·m）



设角速度为 ω (rad/s)

$$P = M\omega \qquad P\times 10^3 = M\times\frac{2\pi n}{60}$$

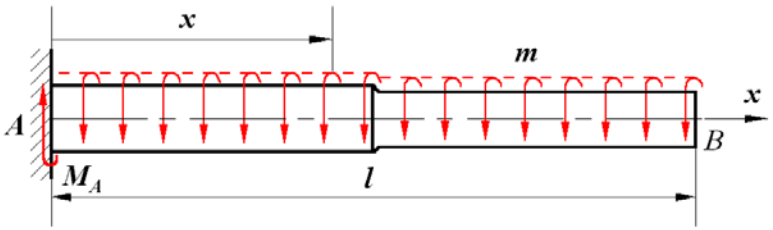
$$\boxed{|M|_{\mathrm{N\cdot m}} = 9549\frac{|P|_{\mathrm{kW}}}{|n|_{\mathrm{r/min}}}}$$

例：$P = 5$ kW，n=1450 r/min，则

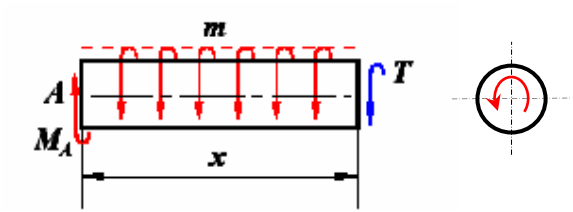
$$M = \left(9540\times\frac{5}{1450}\right)(\mathrm{N\cdot m}) = 32.9\ \mathrm{N\cdot m}$$

3、扭矩与扭矩图

图示阶梯杆件受均匀分布扭力偶矩m作用，m——单位长度上的扭力偶矩



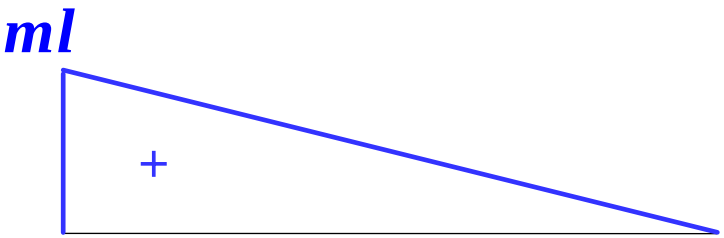
$$M_A = ml$$



截面法求扭矩

分离体平衡

$$T = M_A - mx = m(l - x)$$



扭矩

- 横截面内绕轴线的力矩；
- 矢量沿外法线为正。

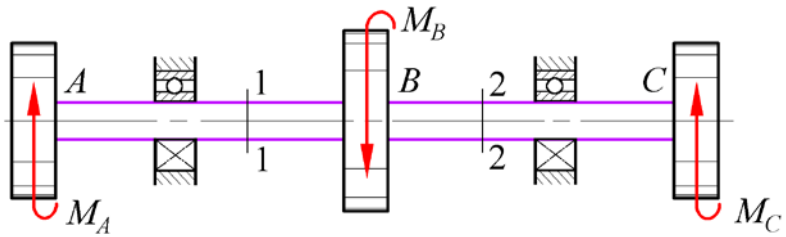
扭矩图

- 各截面扭矩沿轴线的变化；
- 确定最大扭矩所在截面

——危险截面。

4、例题

例4-1　M_A=76 N·m, M_B=191 N·m, M_C=115 N·m，画扭矩图



解：

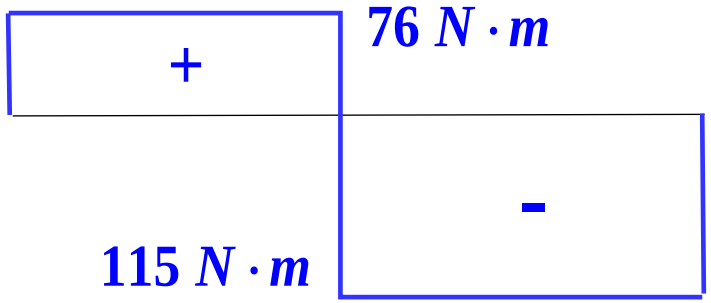
1）分段切开截面，取分离体；

2）分分离体受力分析，扭矩设正；

3）列平衡方程，求扭矩；

$$T_1 = M_A = 76\ \mathrm{N\cdot m}$$

$$T_2 + M_C = 0 \qquad T_2 = -115\ \mathrm{N\cdot m}$$



4）画扭矩图，标注大小、正负，与圆轴对齐。

杆件、分离体、扭矩图对齐

第四章 扭转

4.2 圆轴扭转应力

1、圆轴扭转平面假设

2、圆轴扭转横截面切应力

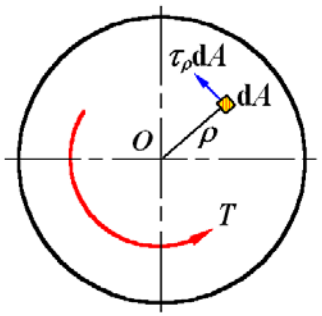
3、薄壁圆管横截面切应力

4、圆轴扭转斜截面应力

1、圆轴扭转平面假设

问题：已知扭矩，求横截面应力 $\int_A \rho\tau_\rho dA = T$

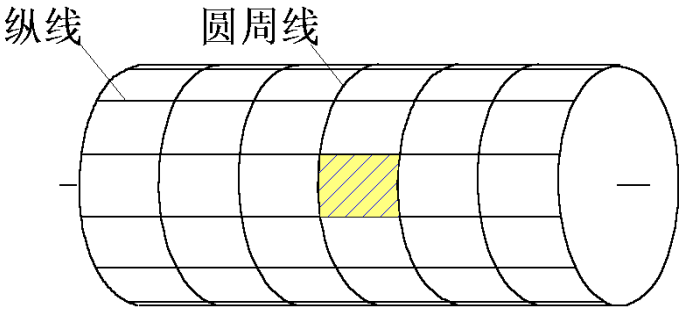
高度静不定



分析思路：

综合考虑静力学、几何、物理三个方面

圆轴扭转表面变形观察



1、圆轴扭转平面假设

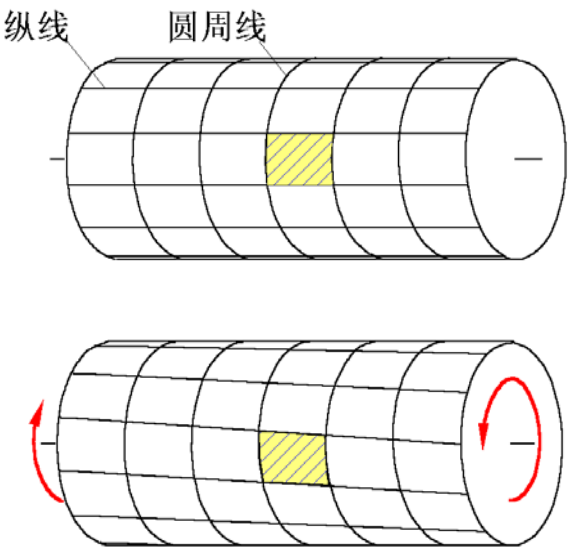
圆轴扭转表面变形观察

圆轴扭转平面假设

各横截面如同刚性平面，
仅绕轴线作相对转动

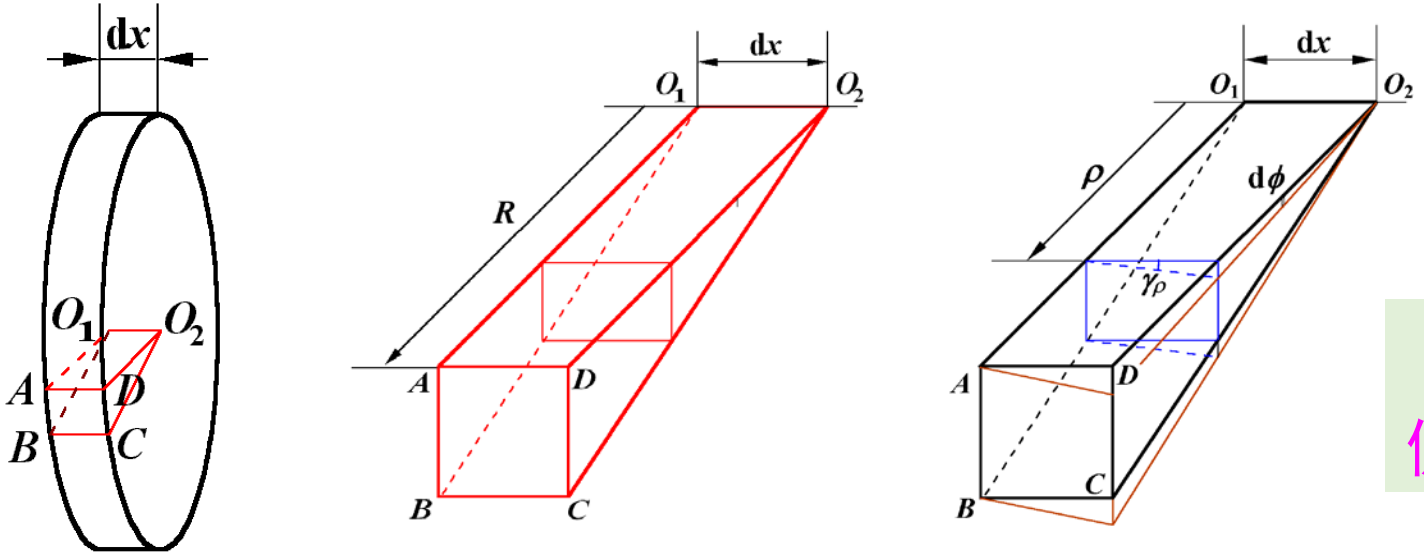
1、圆轴扭转平面假设

圆轴扭转表面变形观察



圆轴扭转平面假设

各横截面如同刚性平面，
仅绕轴线作相对转动



楔形体
侧边错动

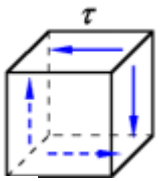
2、圆轴扭转横截面切应力

综合考虑静力学、几何、物理三个方面

几何方面

$$\gamma_\rho \approx \tan\gamma_\rho = \frac{\overline{dd'}}{\overline{ad}} = \rho\frac{\mathrm{d}\varphi}{\mathrm{d}x}$$

仅在垂直半径的平面内有切应变

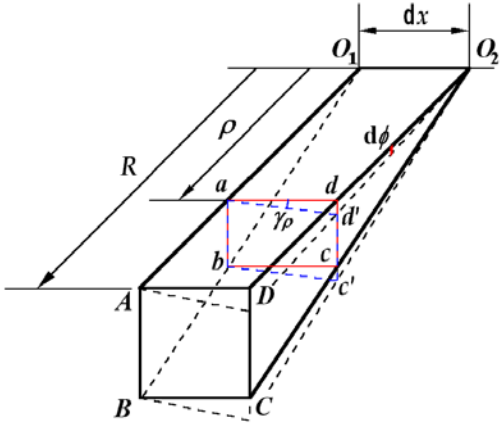


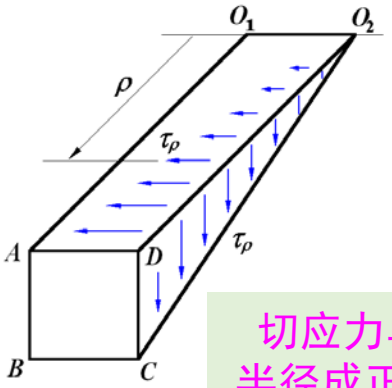
切应力与半径垂直

物理方面

$$\tau_\rho = G\rho\frac{\mathrm{d}\varphi}{\mathrm{d}x}$$

$\mathrm{d}\varphi/\mathrm{d}x$—扭转角变化率





切应力与半径成正比

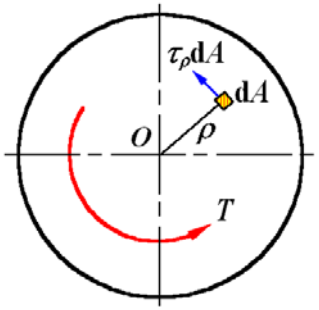
静力学方面

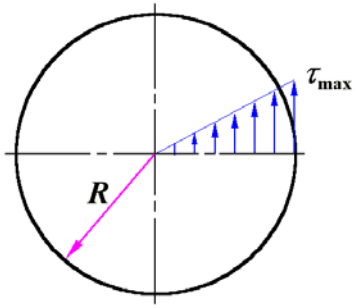
$$\int_A \rho\cdot\tau_\rho \mathrm{d}A = T$$

$$\frac{\mathrm{d}\varphi}{\mathrm{d}x} = \frac{T}{GI_\mathrm{p}} \qquad \tau_\rho = \frac{T\rho}{I_\mathrm{p}} \qquad \tau_{\max} = \frac{T}{W_\mathrm{p}}$$

$I_\mathrm{p} = \int_A \rho^2 \mathrm{d}A$ —极惯性矩

$W_\mathrm{p} = \dfrac{I_\mathrm{p}}{R}$ —抗扭截面模量（系数）





试验表明：适用于圆轴，$\tau_{\max} \le \tau_\mathrm{p}$

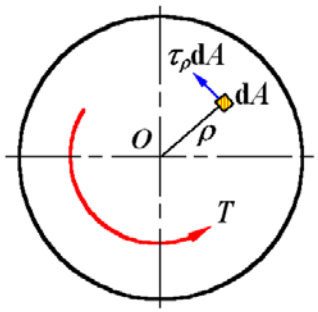
2、圆轴扭转横截面切应力

思考

1）截面合力如何平衡

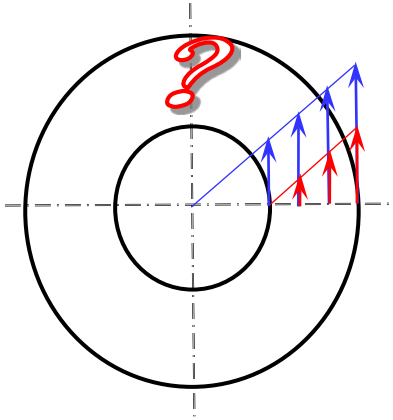
$$\int_A \rho\tau_\rho dA = T$$

$$\int_A \tau_{\rho x} dA = 0,\quad \int_A \tau_{\rho y} dA = 0$$



2）空心圆截面

$$\tau_\rho = \frac{T\rho}{I_p}$$



内径边缘处，
切应力为何不为零？

极惯性矩与抗扭截面系数

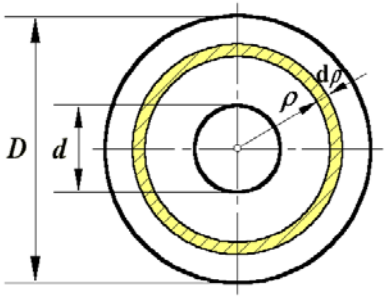
空心圆截面

$$dA = 2\pi\rho d\rho$$

$$I_p = \int_{d/2}^{D/2} \rho^2 \cdot 2\pi\rho d\rho$$

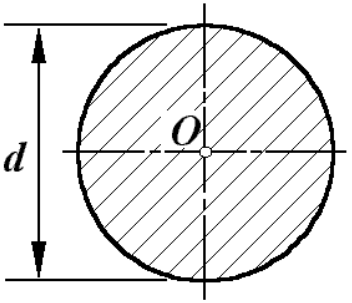
$$= \frac{\pi D^4}{32}\left(1-\alpha^4\right) \qquad W_p = \frac{\pi D^3}{16}\left(1-\alpha^4\right)$$

$$\alpha = \frac{d}{D}$$



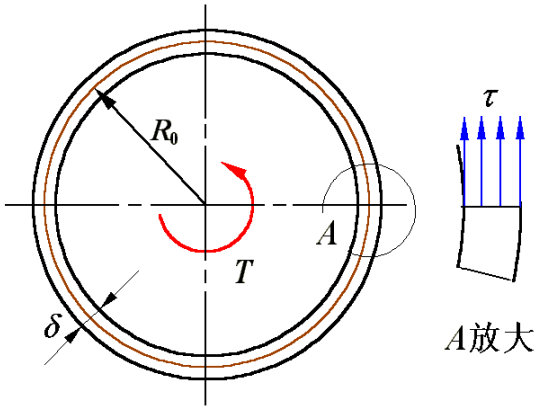
圆截面

$$I_p = \frac{\pi d^4}{32} \qquad W_p = \frac{\pi d^3}{16}$$



3、薄壁圆管横截面切应力

图示薄壁圆管，平均半径R_0，厚度δ，受扭矩T，求横截面切应力。



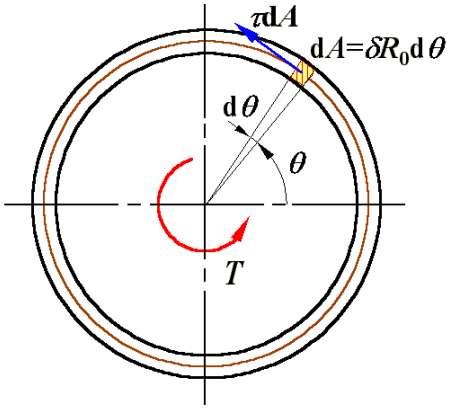
精确解（线弹性情况）

$$\tau_{\max}=\frac{16T}{\pi D^3(1-\alpha^4)} \qquad D=2R_0+\delta,\quad \alpha=\frac{2R_0-\delta}{2R_0+\delta}$$

近似解

假设：1）切应力平行于中心线切线；

2）沿壁厚均匀分布。



切应力轴对称

$$T=\int_0^{2\pi} R_0\cdot\tau\delta R_0\mathrm{d}\theta \Longrightarrow \boxed{\tau=\frac{T}{2\pi R_0^2\delta}=\frac{T}{2\Omega\delta}}$$

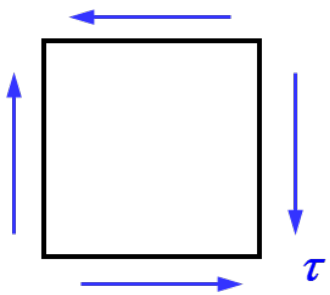
$R_0=10\delta$,

$$\frac{\tau_{\max}-\tau}{\tau_{\max}}=0.045$$

- 适用于所有匀质薄壁杆，包括弹性、非弹性、各向同性与各向异性情况；
- 当$\delta\leqslant R_0/10$时（薄壁圆管），误差≤4.5%.

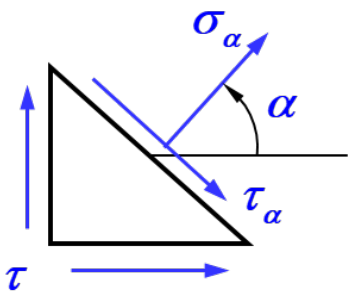
4、圆轴扭转斜截面应力

圆轴横截面上的任一点处于纯剪切状态。



斜截面应力

$$\sigma_\alpha = -\tau \sin 2\alpha, \quad \tau_\alpha = \tau \cos 2\alpha$$

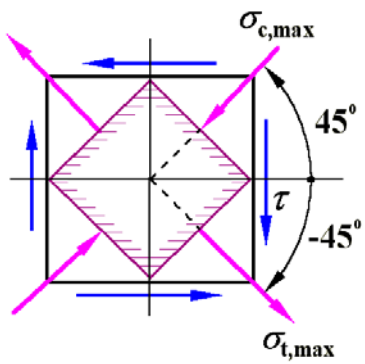


微体
受力平衡

最大应力

$$\alpha = 0, 90^\circ, \quad \tau_{max} = \pm\tau;$$

$$\alpha = \pm 45^\circ, \quad \sigma_{max} = -\sigma_{min} = \tau$$



第四章 扭转

4.3 圆轴扭转强度条件与合理强度设计

1、扭转失效与扭转极限应力

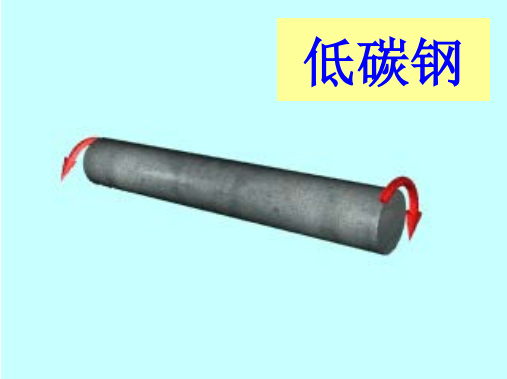
2、圆轴扭转强度条件

3、圆轴合理强度设计

4、例题

1、扭转失效与扭转极限应力

圆轴扭转试验



试验表明：

- 线弹性： $\tau = G\gamma$ ，当 $\tau < \tau_p$ 剪切比例极限；
- 屈服：纵、圆周向有滑移线，扭转屈服应力 τ_s ；
- 断裂：横截面，扭转强度极限 τ_b 。

- 切应力与切应变近似线弹性；
- 断裂：45° 螺旋面，扭转强度极限 τ_b。

$$\tau_s, \tau_b \Rightarrow \tau_u$$ 扭转极限应力

2、圆轴扭转强度条件

$$\tau_{max} \le [\tau]$$

$$[\tau] = \frac{\tau_u}{n}$$

变截面圆轴：

$$\tau_{max} = \left(\frac{T}{W_p}\right)_{max}$$

等截面圆轴：

$$\tau_{max} = \frac{T_{max}}{W_p}$$

τ_u—材料的扭转极限应力

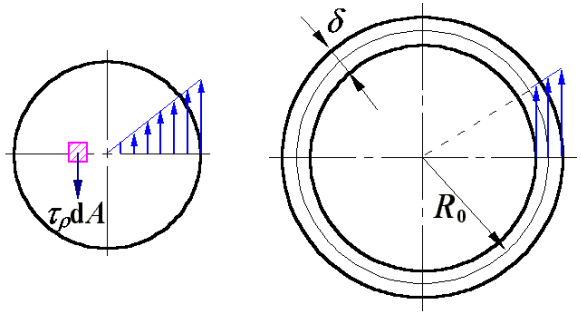
n —安全因数

塑性材料：$[\tau]$ = （0.5~0.577）$[\sigma]$

脆性材料：$[\tau]$ = （0.8~1.0）$[\sigma_t]$

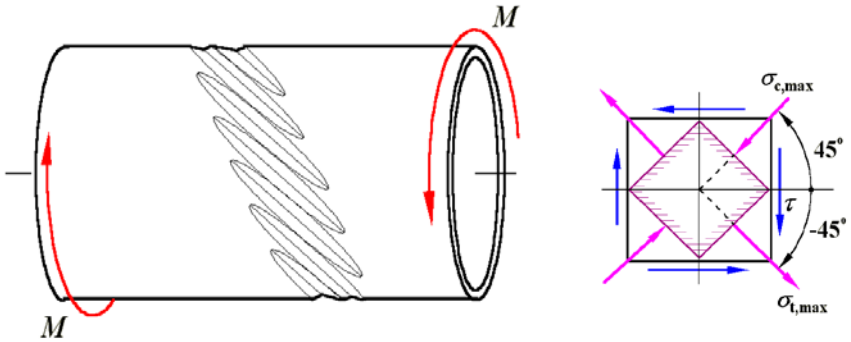
3、圆轴合理强度设计

1）合理截面形状



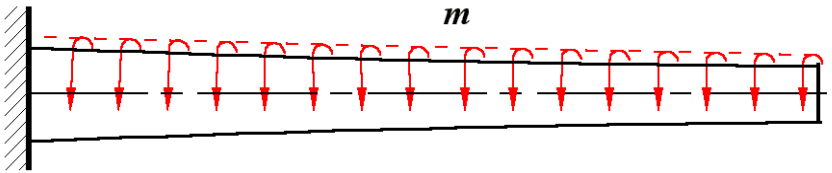
空心截面好

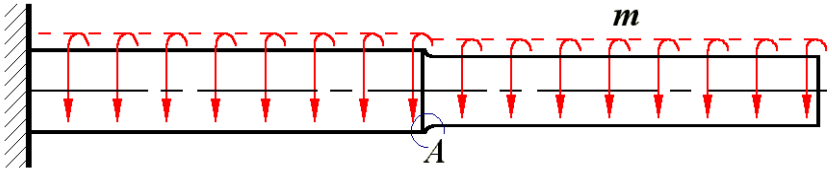
材料承载效率高

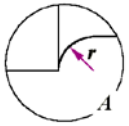


过薄会失稳，要统筹兼顾

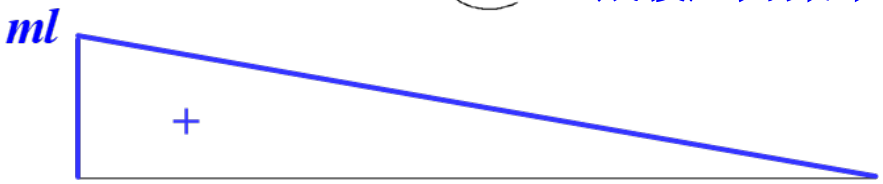
2）采用变截面轴与阶梯形轴







减缓应力集中



4、例题

例4-2 已知 $T=1.5$ kN·m，$[\tau]=50$ MPa。

试根据强度条件设计实心圆轴与 $\alpha=0.9$ 的空心圆轴。

解： $\tau_{\max}=\dfrac{T}{W_p}\le[\tau]$，$W_{p1}=\dfrac{\pi d^3}{16}$，$W_{p2}=\dfrac{\pi D^3}{16}\left(1-\alpha^4\right)$

1）确定轴径

$$d\ge\sqrt[3]{\frac{16T}{\pi[\tau]}}=\sqrt[3]{\frac{16\times(1.5\times10^6\,\mathrm{N\cdot mm})}{\pi\times(50\mathrm{MPa})}}$$

$$=53.5\ \mathrm{mm}$$

$$D\ge\sqrt[3]{\frac{16T}{\pi(1-\alpha^4)[\tau]}}=76.3\ \mathrm{mm}$$

取： $d=54\ mm$， $D=77\ mm$.

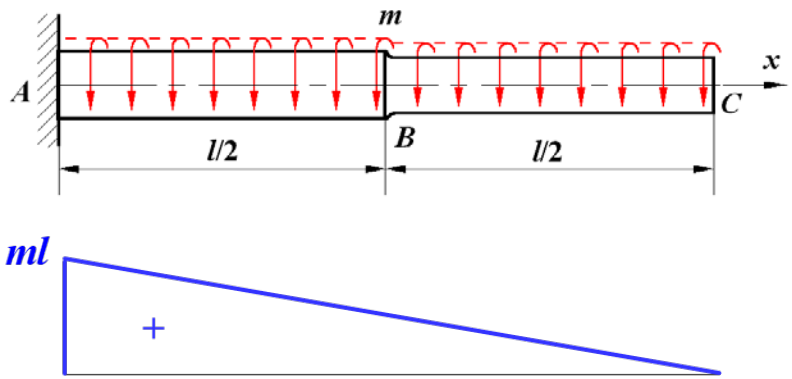
2）重量比较

$$\beta=\frac{\frac{\pi}{4}D^2(1-\alpha^2)}{\frac{\pi}{4}d^2}=39.5\%$$

空心轴远比实心轴轻

4、例题

例4-3 图示阶梯型圆轴，左、右段直径分别为 $d_1 = 75$ mm，$d_2 = 60$ mm，$l = 1$ m，$[\tau] = 50$ MPa，受均布扭力矩 $m = 3500$ N·m/m，试校核强度。



解：1）扭矩分析 ➡ 危险截面

截面 A：大扭矩；

截面 B：小截面。

可能有最大应力

2）逐一计算，校核强度

$$\tau_A = \frac{16T_A}{\pi d_1^3} = \frac{16\times 3500\times 10^3}{3.14\times 75^3} = 42.3\ \text{MPa} < [\tau]$$

$$\tau_B = \frac{16T_B}{\pi d_2^3} = \frac{16\times 1750\times 10^3}{3.14\times 60^3} = 41.3\ \text{MPa} < [\tau]$$

圆轴强度足够

4、例题

例4-4 图示阶梯型薄圆管，平均半径均为R_0，已知R_0, M_e, [τ]，$l_1+l_2=l$。问：两段长度和厚度l_1, t_1和l_2，t_2为多少时，圆管最轻。

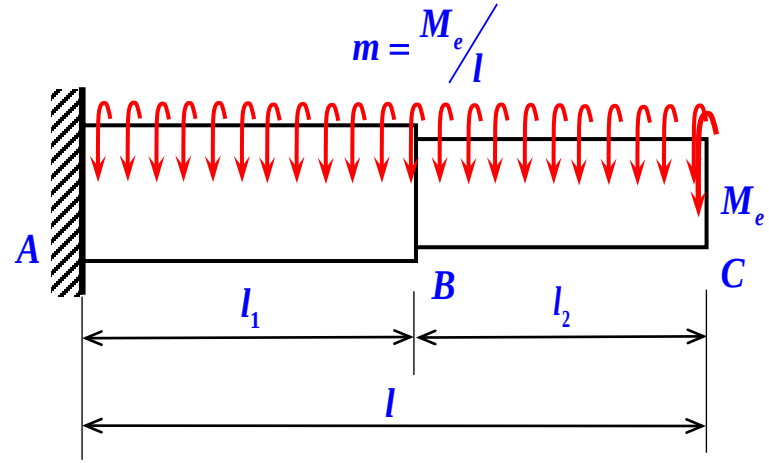
解： 1）强度分析

危险截面：A、B

$$T_A = 2M_e,\ T_B = M_e\left(1+\frac{l_2}{l}\right)$$

$$\begin{cases} \dfrac{T_A}{2\pi R_0^2 t_1} = [\tau] \\ \dfrac{T_B}{2\pi R_0^2 t_2} = [\tau] \end{cases} \Rightarrow \begin{cases} t_1 = \dfrac{M_e}{\pi R_0^2 [\tau]} \\ t_2 = \dfrac{M_e}{2\pi R_0^2 [\tau]}\left(1+\dfrac{l_2}{l}\right) < t_1 \end{cases}$$

等强情况无数
需讨论极值



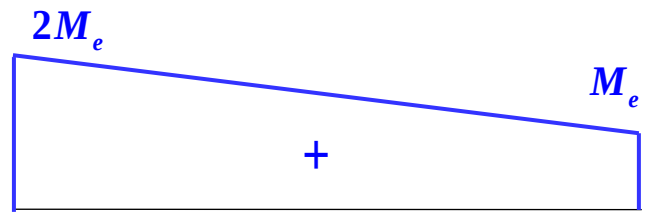
2）重量分析

$$W = \gamma 2\pi R_0 (t_1 l_1 + t_2 l_2)$$

$$= \frac{\gamma M_e l}{R_0 [\tau]}\left\{2 - \frac{l_2}{l} + \left(\frac{l_2}{l}\right)^2\right\}$$

$$\frac{dW}{dl_2} = 0 \Rightarrow \begin{cases} l_2 = l_1 = \dfrac{l}{2} \\ t_2 = \dfrac{3M_e}{4\pi R_0^2 [\tau]} \end{cases}$$

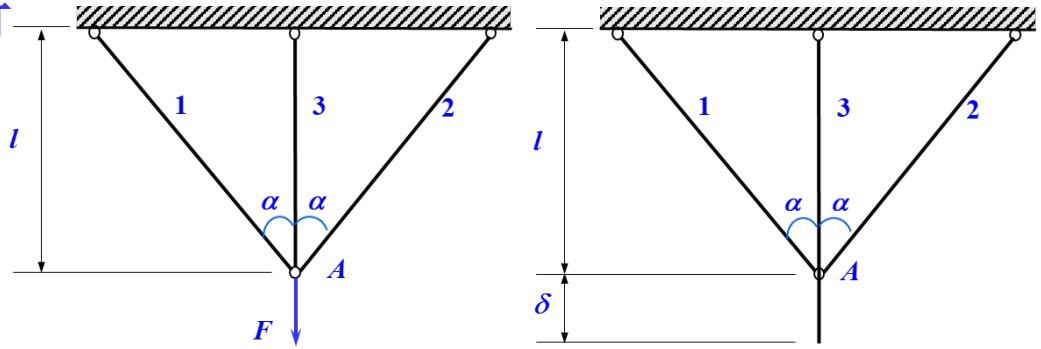
$$\frac{d^2W}{dl_2^2} > 0$$



等强之下
再取重量极小

3、制造偏差引起的杆长变化

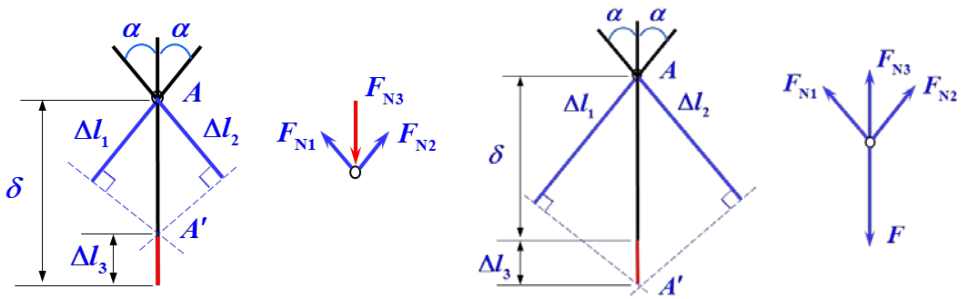
例3-9 图示对称静不定桁架，三杆*EA*相同，竖杆受力偏大，可否改进？（习题3-30）



解：问题分析

竖杆短、且变形大，故受力大。

使竖杆超长 δ



装配后的变形与受力　　加载后的变形与受力

- 加载前，竖杆受压、斜杆受拉；
- 加载后，斜杆继续受拉，竖杆先卸载、再加载，受力变小。

看清楚变形图中的制造偏差

本讲：圆轴扭转应力

圆轴扭转平面假设

下讲：圆轴扭转变形

与非圆截面杆件扭转